Architecture

Data flow



1. Ingestion

- Gin HTTP router
  > Access to a url path via web app.
- RBAC (Role-based access control) middleware
  > Controls permissions for every idea, keyword, and action according to the user role to enforce enterprise privacy.

2. Validation

- Go input type validation handler
  > Validates that submitted files (.pdf, .csv, .txt, .docx) match allowed extensions and size limits before processing.

3. Preprocessing

- Go chunkenizer
  > Splits long documents into manageable, coherent text segments so the local model can ingest them within its context limits.
- Go structs
  > Type-safe memory structures that hold and transform validated raw data across backend layers.
- SQLC payload validation
  > Validates and coerces incoming request parameters into generated database types to reject malformed data before hitting the database.

4. Async dispatch

- Go buffered channel

> Queues validated document tasks in memory so the HTTP handler can respond immediately without blocking the client.

- Bounded worker pool
  > Limits the number of concurrent background tasks to prevent local CPU and memory exhaustion during heavy inference.

5. Semantic extraction

- llama server
  > High-performance C++ inference engine that exposes a lightweight local HTTP endpoint for the model without Python overhead.
- Qwen 2.5 3B GGUF
  > Fast, low-footprint local language model that reads the text to extract summaries, entities, and conceptual relations.
- Grammar JSON
  > Forces the model output into an exact schema via GBNF grammars to ensure deterministic parsing in Go.

6. Vector embedding

- specialized embedding model
  > Translates extracted concepts and idea texts into dense numerical vectors that capture geometric semantic meaning.
- PGVector vector cords storage and indexation
  > Stores embeddings in PostgreSQL and accelerates similarity searches through HNSW indexing.

7. Linking

- Postgres knn search
  > Queries the nearest neighbor vectors in the database to identify candidate ideas with high semantic similarity.
- Go overlap concept linker
  > Computes intersections between extracted entity tags to reinforce edges that share common named concepts.
- Go bidirectional injector

> Creates reciprocal graph edges in the database and inserts bidirectional links into markdown documents.

8. Wiki and consulting

- Go graph traversal PPR
  > Executes Personalized PageRank walks across graph edges to retrieve deep, multi-hop related ideas without flat chunking.
- Gin HTML templates
  > Renders server-side web layouts and interactive graph views directly to the browser.
- Postgres knowledge graph
  > The persistent relational store that preserves all nodes, edges, and permission constraints as a compounding single source of truth.